REGRESSÃO

$$Y = f(X)$$

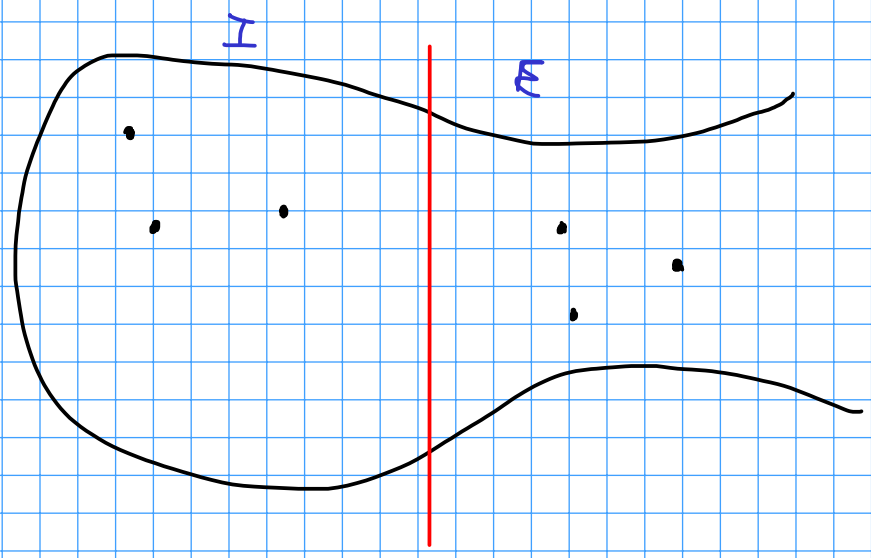
Y → NUM X → NUM

ANOVA

$$Y = f(X)$$

Y → NUM X → CATEGÓRICA (FATOR)

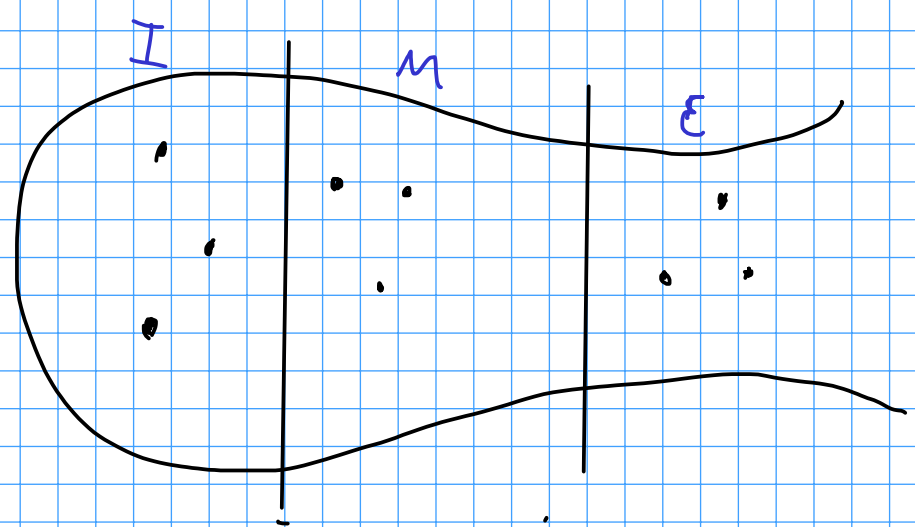
ANOVA c/ + DE 1 FATOR



FATOR ZONA c/ 2 NÍVEIS: I, E

$$\begin{cases} H_0: \mu_E = \mu_I \\ H_1: \mu_E \neq \mu_I \end{cases}$$

Teste-t

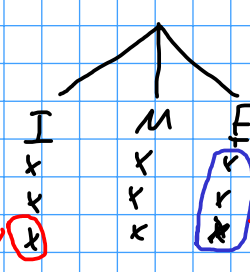


FATOR ZONA c/ 3 NÍVEIS: I, M, E

$$\begin{cases} H_0: \mu_E = \mu_M = \mu_I \\ H_1: \text{PELO MENOS 1 É} \neq \end{cases}$$

ANOVA

PSEUDOREPLICAÇÃO = CONFUNDIMENTO ESPACIAL



→ FATOR ZONA (TRAT. EXP.)

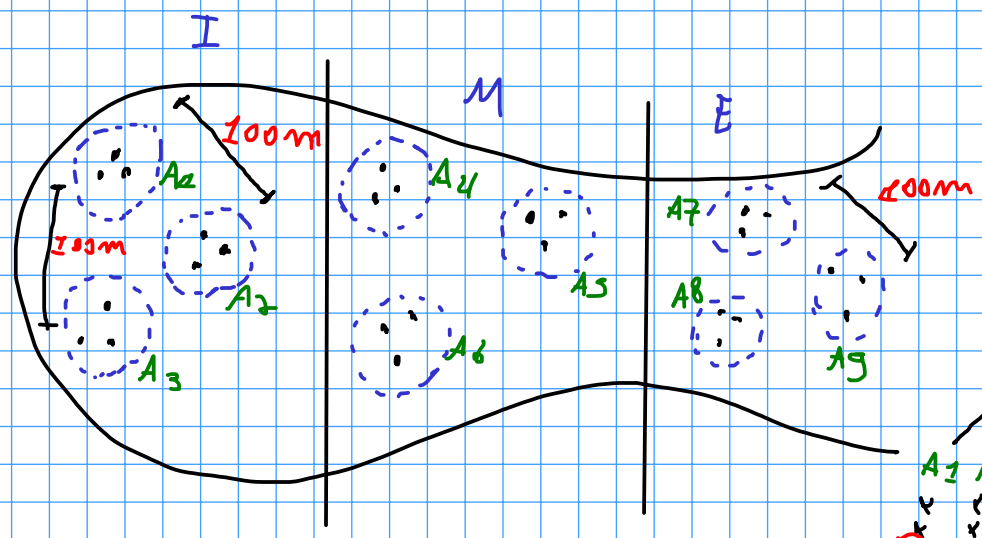
→ VARIABILIDADE DO RESÍDUO

$$X = \mu + \tau + \varepsilon$$

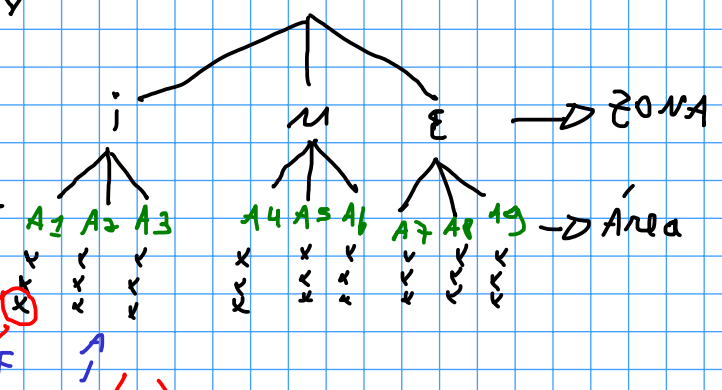
↓

UNIFATORIAL

	SQ	G.L	M.Q	F
ZONA				$M.Q_{ZONA} / MQ_{RES}$
RESÍDUO OU ERRO				



RESOLVE A PSEUDOREPLICAÇÃO ESPACIAL INCLUINDO UMA ESCALA INTERMEDIÁRIA DE VARIABILIDADE



$$X = \mu + Z + A(Z) + e$$

HIERÁRQUICO OU NESTED OU ANINHADO

	S.Q.	G.L.	M.Q.	F
Z				$MQ_Z / MQ_{A(Z)}$
$A(Z)$				$MQ_{A(Z)} / MQ_e$
e				

HIPÓTESES (UMA P/ CADA TERMO)

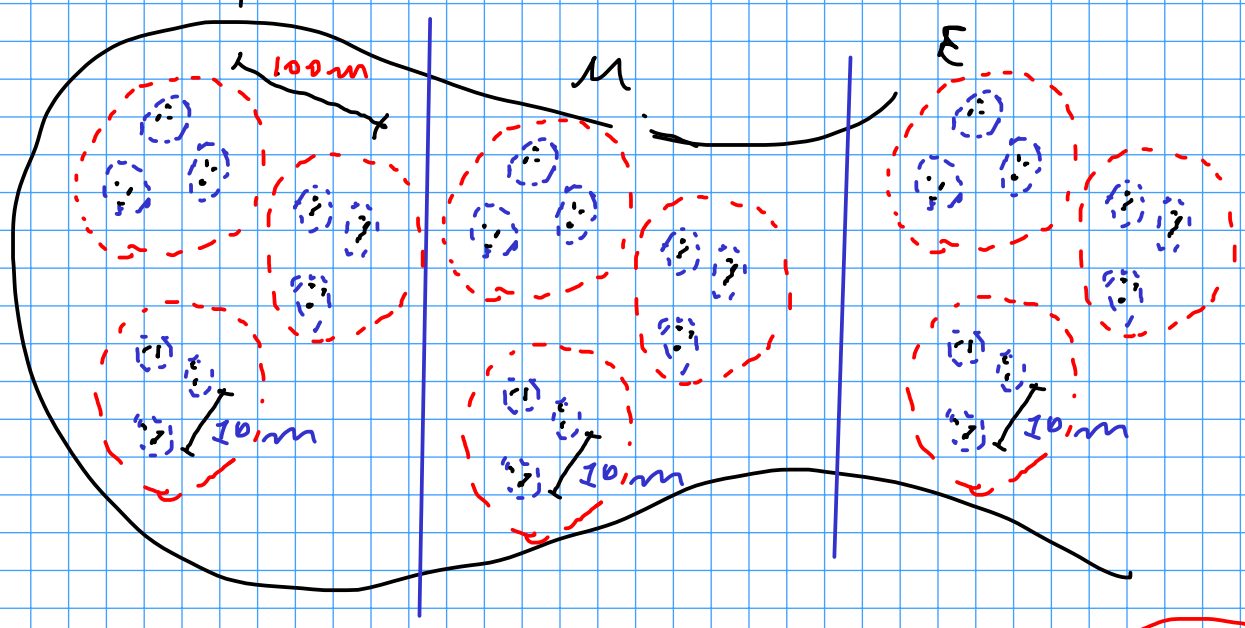
- PARA ZONAS $\begin{cases} H_0: \mu_I = \mu_M = \mu_E \\ H_a: \text{PELO MENOS 1 DELAS É DIFERENTE} \end{cases}$

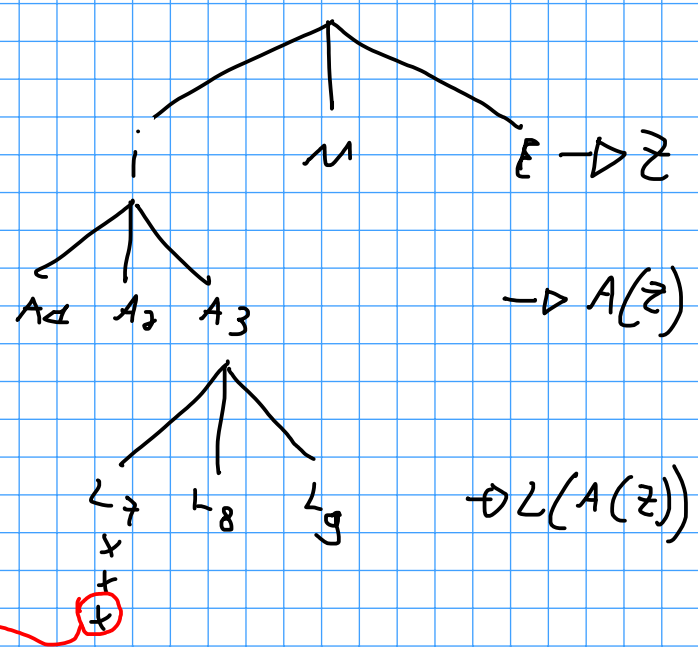
- PARA ÁREAS

  ↳ DEPENDE DA ZONA CONSIDERADA

$$\begin{cases} H_0: I: \mu_{A_1} = \mu_{A_2} = \mu_3 \\ M: \mu_{A_4} = \mu_{A_5} = \mu_{A_6} \\ E: \mu_{A_7} = \mu_{A_8} = \mu_{A_9} \end{cases}$$

3 FATORES HIERARQUICOS





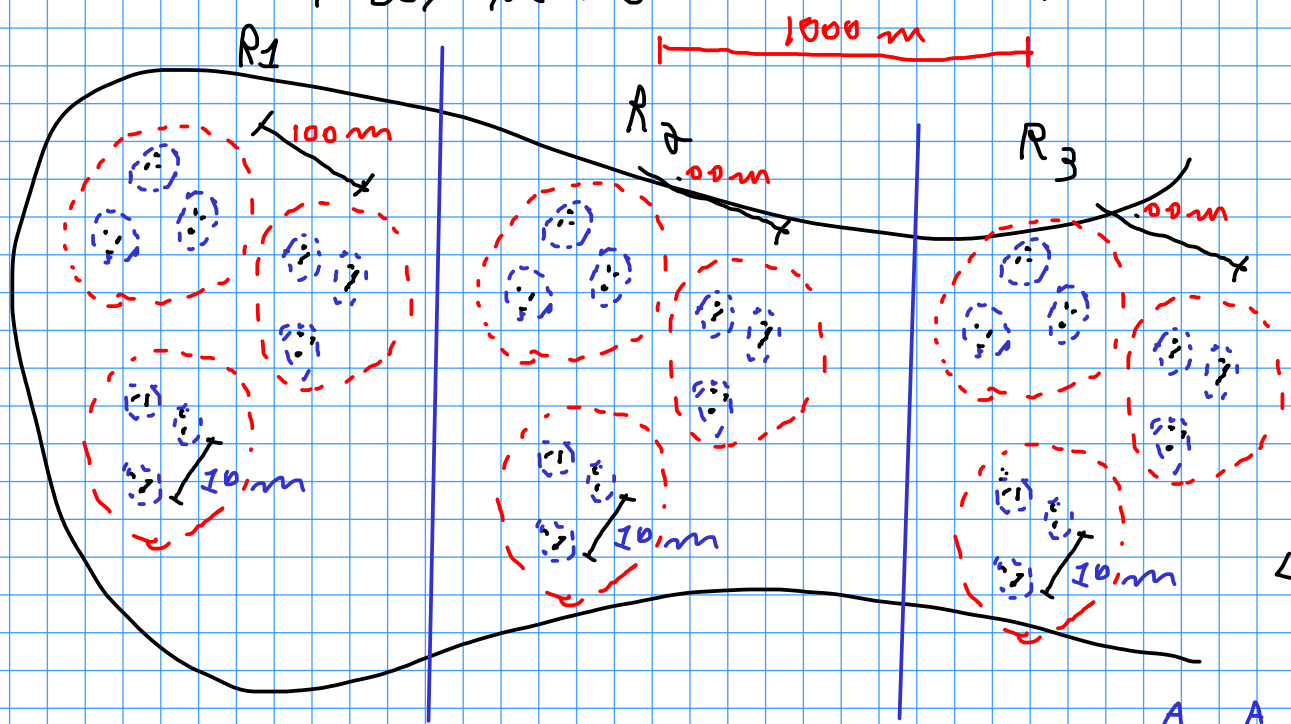
$$3Z \times 3A \times 3L \times 3R = 81$$

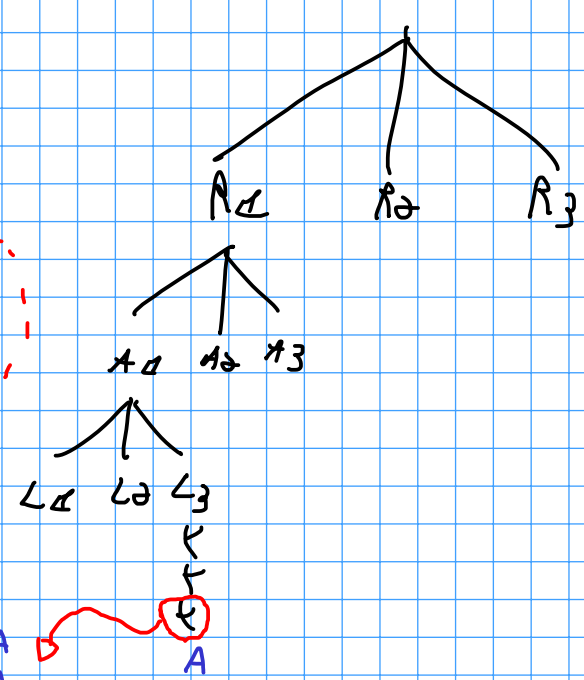
$$X = \mu + Z + A(Z) + L(A(Z)) + e$$

	SQ	G.L	M.Q.	F
Z				$MQ_Z / MQ_{A(Z)}$
A(Z)				$MQ_{A(Z)} / MQ_{L(A(Z))}$
L(A(Z))				$MQ_{L(A(Z))} / MQ_e$
e				

DELINEAMENTOS EM ZOOM SÃO CHAMADOS DE FULLY NESTED

FULLY NESTED S/ TRATAMENTO EXPERIMENTAL





$$X = \mu + R + A(R) + L(A(R)) + e$$

				F
R				$MQ_R / MQ_{A(R)}$
A(R)				⋮
L(A(R))				⋮
e				

EXEMPLOS EM OUTRAS ÁREAS

- MEDICINA → TESTE DE UM MEDICAMENTO

PLACEBO MEDICAM. → Teste-t

PLACEBO MEDICAM. → Trat. Exp. (F)

→ COBAIAS (Trat) (A)

C1 C2 C3 C4 C5 C6

- SOCIAL

PÚBLICA PARTICULARES → Trat. Exp. (F)

→ ESCOL (A)

→ PROF. (A)

E1 E2 E3 E4 E5 E6

P1 P2 P3

POLÍTICAS DE MANEJO

POL 1 POL 2 POL 3 → Trat. Exp. (F)

→ UNIDADES (A)

UC1 UC2 UC3 UC4 UC5 UC6